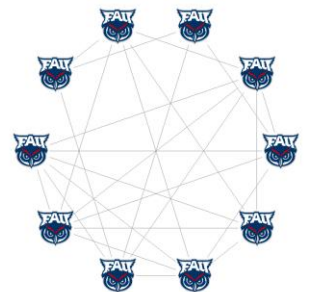




Introduction to Network Analysis

Dr. Tom R. Leppard

github.com/Tom-R-Leppard/SNA_Workshop



1: Install R

RStudio requires R 3.6.0+. Choose a version of R that matches your computer's operating system.

R is not a Posit product. By clicking on the link below to download and install R, you are leaving the Posit website. Posit disclaims any obligations and all liability with respect to R and the R website.

DOWNLOAD AND INSTALL R

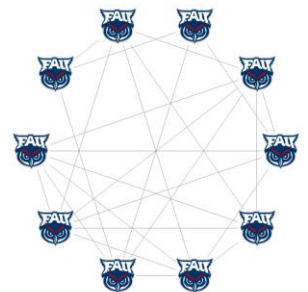
2: Install RStudio

DOWNLOAD RSTUDIO DESKTOP FOR WINDOWS

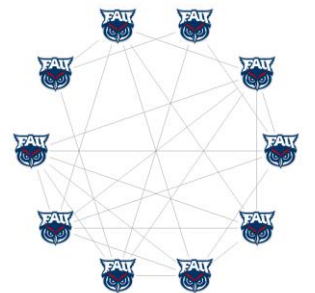
Size: 287.97 MB | [SHA-256: 8CE88C63](#) | Version: 2025.09.0+387 | Released: 2025-09-12

R and R Studio – Download NOW

<https://posit.co/download/rstudio-desktop/>

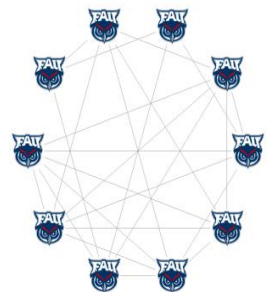


Let's Meet

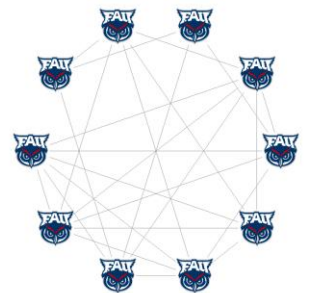


What are Networks? What is Network Analysis?

- A system with individual interacting parts
- A visual representation of a system or group of people
 - Social, transportation, neural, epidemiological, cellular, commercial (travel/trade).
- A process of analysing individuals, groups, and/or systems using graph theory
 - Analyse observed, latent, and structural social processes.
- Building statistical models to explain why interactions occur, information spreads, or the prominence of people.

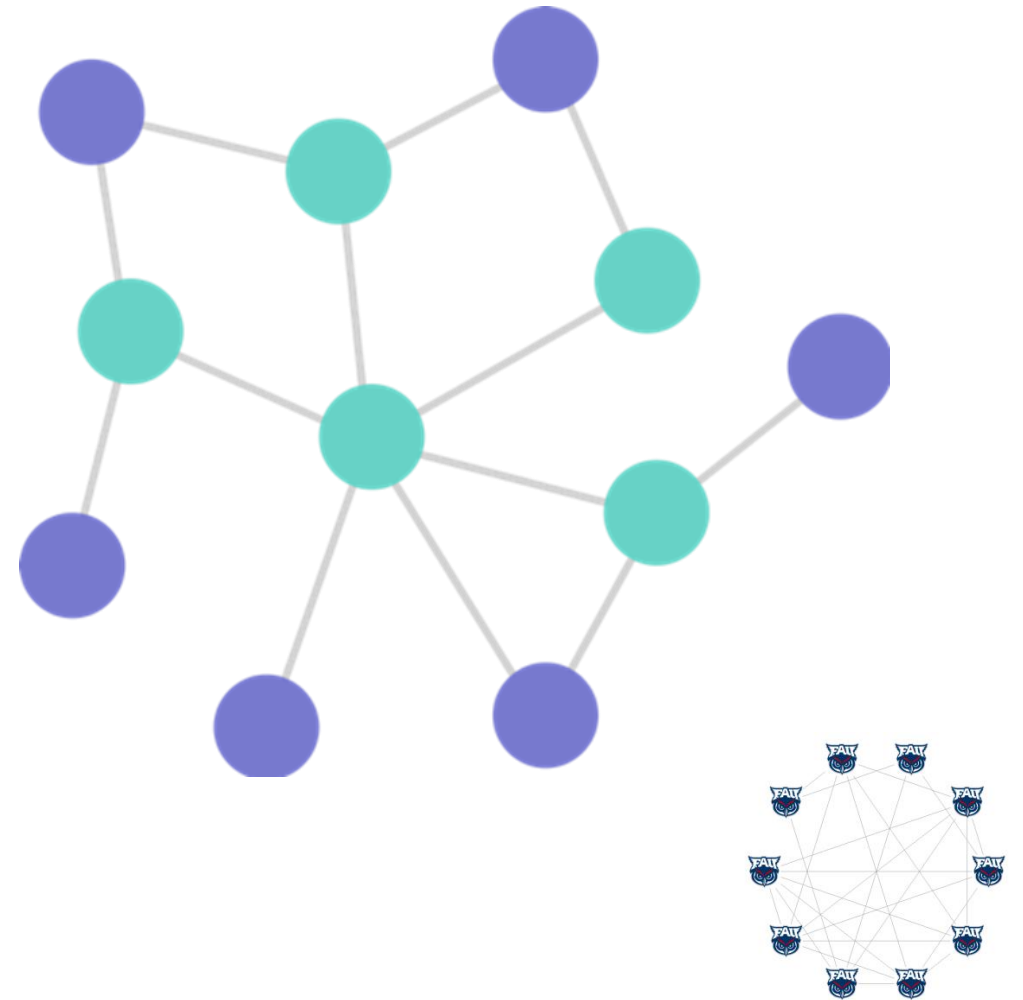


Network Components



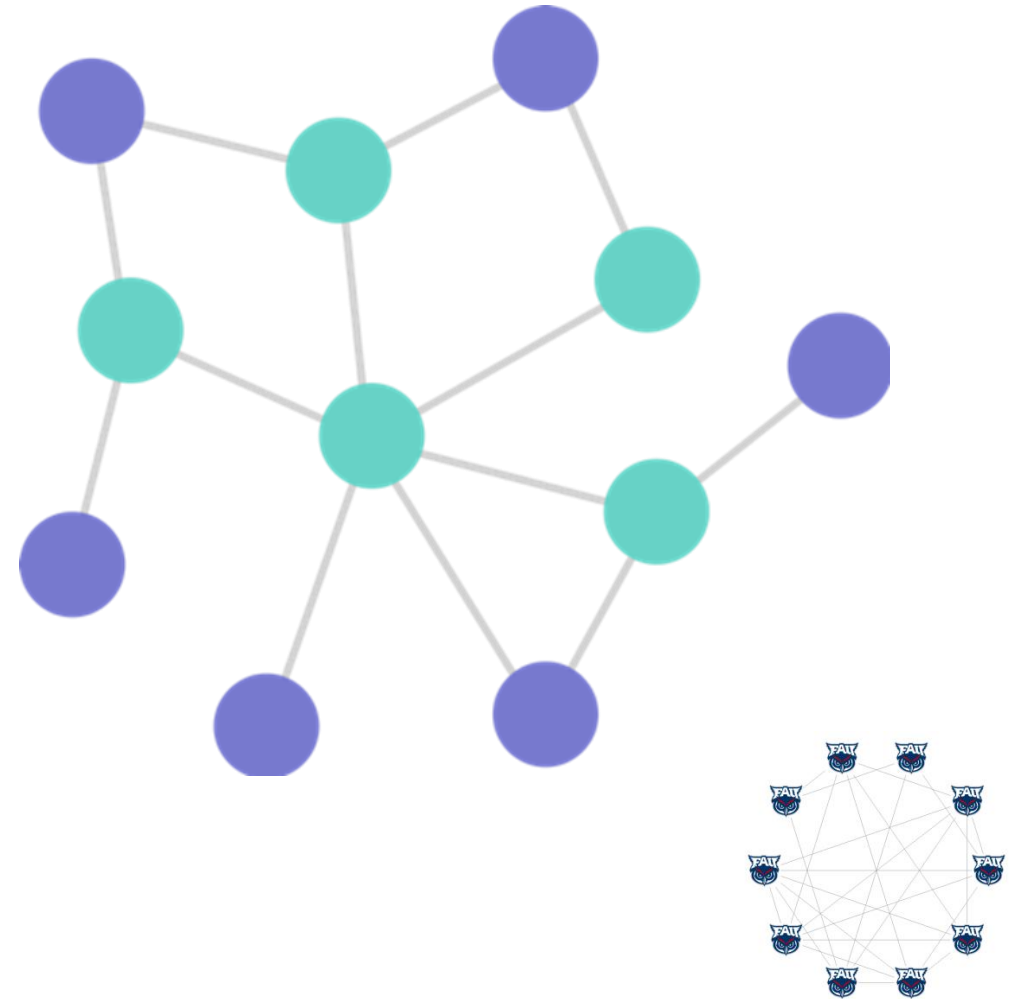
Components - Nodes

- The nodes are the actors of a network.
 - People, groups, organizations, states, words, sentences, books, etc.
- The nodes in the figure are represented by the purple and green dots.

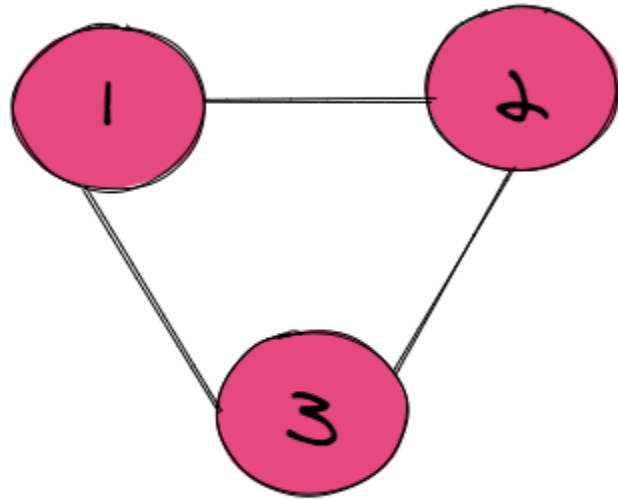


Components - Edges

- Edges are the connections in the network.
 - Kinship, affiliation, interaction, affective relationships, etc.
- The edges are represented in the figure by the gray lines linking each node to another.



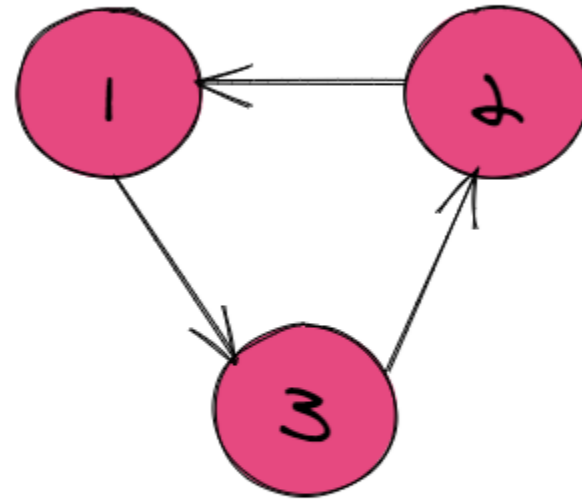
Undirected graph



Vertices = $\{1, 2, 3\}$

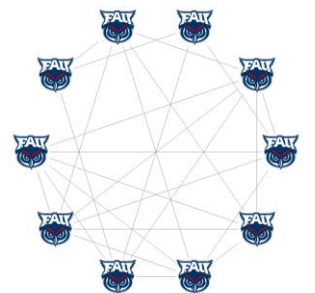
Edges = $\{\{1, 2\}, \{2, 3\}, \{3, 1\}\}$

Directed graph

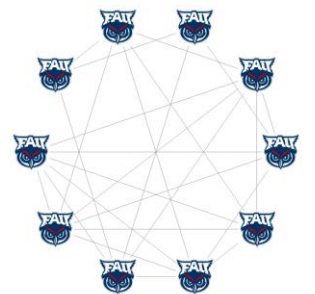


Vertices = $\{1, 2, 3\}$

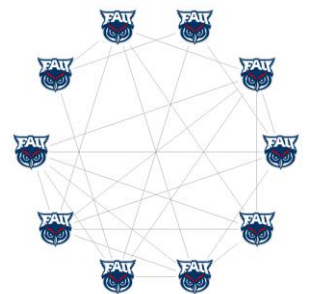
Edges = $\{\{2, 1\}, \{1, 3\}, \{3, 2\}\}$



GO TO RSTUDIO



BE FREE 😊



Similarities			Social Relations				Interactions	Flows
Location e.g., Same spatial and temporal space	Membership e.g., Same clubs Same events etc.	Attribute e.g., Same gender Same attitude etc.	Kinship e.g., Mother of Sibling of	Other role e.g., Friend of Boss of Student of Competitor of	Affective e.g., Likes Hates etc.	Cognitive e.g., Knows Knows about Sees as happy etc.	e.g., Sex with Talked to Advice to Helped Harmed etc.	e.g., Information Beliefs Personnel Resources etc.

Fig. 3. A typology of ties studied in social network analysis.